

第13讲. Γ 函数. = 重积分一. Γ 函数.

$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx \quad \text{定义域 } (0, +\infty).$$

复变: 设 $f(x, u) \in C(D)$. $D = [a, +\infty) \times [\alpha, \beta]$. $\therefore \varphi(u) = \int_a^{+\infty} f(x, u) dx$ 关于 $u \in [\alpha, \beta]$ 一致收敛.

$$① \varphi(u) \in C[\alpha, \beta].$$

$$② \varphi(u) \in R[\alpha, \beta] \quad \text{且} \quad \int_{\alpha}^{\beta} \left(\int_a^{+\infty} f(x, u) dx \right) du = \int_a^{+\infty} \left(\int_{\alpha}^{\beta} f(x, u) du \right) dx$$

$$③ \forall u \in [\alpha, \beta]. \quad \int_a^{+\infty} f(x, u) dx \text{ 收敛. 且 } f'_u(x, u) \in C(D).$$

$$\int_a^{+\infty} f'_u(x, u) dx \text{ 关于 } u \in [\alpha, \beta] \text{ 上一致收敛.}$$

$$④ \varphi(u) = \int_a^{+\infty} f(x, u) dx \in C'[\alpha, \beta]. \quad \text{且 } \varphi'(u) = \int_a^{+\infty} f'_u(x, u) dx$$

性质1. $\Gamma(\alpha) \in C(0, +\infty)$. 且在 $(0, +\infty)$ 存在所可导.

$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx = \int_0^1 x^{\alpha-1} e^{-x} dx + \int_1^{+\infty} x^{\alpha-1} e^{-x} dx.$$

$$\forall u_0 \in (0, +\infty). \quad \exists [a, b] \subset (0, +\infty). \text{ s.t. } u_0 \in [a, b].$$

$$D_1 = (0, +1] \times [\alpha, \beta].$$

$$D_2 = [1, +\infty) \times [\alpha, \beta].$$

$$\text{于是 } \int_0^1 x^{\alpha-1} e^{-x} dx \text{ 关于 } \alpha \in [a, b] \text{ 一致收敛.}$$

$$\forall x \in (0, 1]. \quad x^{\alpha-1} e^{-x} \leq x^{\alpha-1} e^{-x}$$

$$\text{于是 } \int_0^1 x^{\alpha-1} e^{-x} dx \text{ 收敛}$$

$$\lim_{x \rightarrow 0^+} x^p \cdot x^{\alpha-1} e^{-x} = 0$$

$$p = 1 - \alpha < 1$$

$$\text{于是 } \int_1^{+\infty} x^{\alpha-1} e^{-x} dx \text{ 关于 } \alpha \in [a, b] \text{ 上一致收敛.}$$

$$x \geq 1. \quad x^{\alpha-1} e^{-x} \leq x^{b-1} e^{-x}$$

$$\text{于是 } \int_1^{+\infty} x^{b-1} e^{-x} dx \text{ 收敛}$$

$$\lim_{x \rightarrow +\infty} \frac{x^p \cdot x^{b+1}}{e^x} = 0$$

$p > 1$

(2) $\forall u_0 \in (0, +\infty)$. 要证 $\Gamma(u)$ 在 u_0 可微. $u_0 \in [a, b]$.

$$\int_0^1 \underline{x^{\alpha+1} e^{-x} \ln x} dx + \int_1^{+\infty} x^{\alpha+1} e^{-x} \ln x dx.$$

证 $\int_0^1 \underline{x^{\alpha+1} e^{-x} \ln x} dx$ 关于 $\alpha \in [a, b]$ 一致收敛.

$x \in (0, 1]$.

$$x^{\alpha+1} e^{-x} |\ln x| \leq x^{\alpha+1} |\ln x| \leq x^{\alpha+1} |\ln x|$$

证 $\int_0^1 x^{\alpha+1} |\ln x| dx$ 收敛.

$$a > 0. \quad \lim_{x \rightarrow 0^+} x^{1-\frac{a}{2}} x^{a-1} |\ln x| = \lim_{x \rightarrow 0^+} \frac{|\ln x|}{x^{-\frac{a}{2}}} = 0.$$

$$p = 1 - \frac{a}{2} < 1.$$

证 $\int_1^{+\infty} x^{\alpha+1} e^{-x} \ln x dx$ 关于 $\alpha \in [a, b]$ 一致收敛.

$$x \geq 1. \quad x^{\alpha} \cdot e^{-x} \frac{\ln x}{x} \leq x^b e^{-x}.$$

$$\int_1^{+\infty} x^b e^{-x} dx \text{ 收敛.}$$

$$\therefore \Gamma'(\alpha) = \int_0^{+\infty} \underline{x^{\alpha+1} e^{-x} \ln x} dx.$$

$$\Gamma^{(n)}(\alpha) = \int_0^{+\infty} x^{\alpha+1} e^{-x} (\ln x)^n dx. \in C(0, +\infty).$$

$$2. \quad \Gamma(\alpha+1) = \alpha \Gamma(\alpha). \quad \alpha > 0.$$

$$\Gamma(\alpha+1) = \int_0^{+\infty} \underbrace{x^{\alpha} e^{-x}}_1 dx = - \underbrace{x^{\alpha} e^{-x}}_0 \Big|_0^{+\infty} + \alpha \int_0^{+\infty} x^{\alpha-1} e^{-x} dx = \alpha \Gamma(\alpha).$$

$$\forall \alpha > 0. \quad \exists n \in \mathbb{N}^+. \text{ s.t. } n < \alpha \leq n+1 \Rightarrow 0 < \alpha - n \leq 1$$

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha) = \alpha(\alpha-1) \Gamma(\alpha-1) = \alpha(\alpha-1) \cdots (\alpha-n) \Gamma(\alpha-n)$$

$$\Gamma(n+1) = n! \Gamma(1)$$

$$\Gamma(n+1) = n! = \int_0^{+\infty} x^n e^{-x} dx$$

$$\Gamma(1) = \int_0^{+\infty} e^{-x} dx = -e^{-x} \Big|_0^{+\infty} = 1.$$

3.

$$\lim_{\alpha \rightarrow +\infty} \Gamma(\alpha) = +\infty.$$

$$\alpha \rightarrow +\infty$$

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha)$$

$$\lim_{\alpha \rightarrow 0^+} \Gamma(\alpha) = \lim_{\alpha \rightarrow 0^+} \frac{\Gamma(\alpha+1)}{\alpha} = +\infty.$$

$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx$$

$$\Gamma'(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} \ln x dx.$$

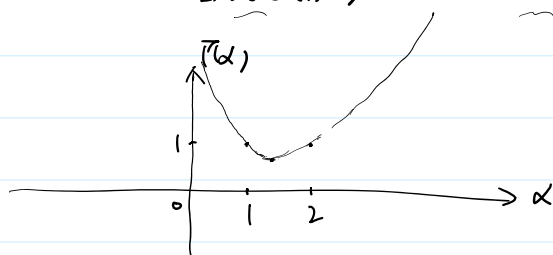
$$= \int_0^1 x^{\alpha-1} e^{-x} \ln x dx + \int_1^{+\infty} x^{\alpha-1} e^{-x} \ln x dx$$

$$\Gamma''(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} (\ln x)^2 dx > 0.$$

$$\Gamma(1) = \int_0^{+\infty} e^{-x} dx = 1$$

$$\Gamma(2) = \int_0^{+\infty} x e^{-x} dx = -x e^{-x} \Big|_0^{+\infty} + \int_0^{+\infty} e^{-x} dx = 1.$$

$$[1, 2] \quad \exists \alpha_0 \in (1, 2) \text{ s.t. } \Gamma'(\alpha_0) = 0.$$



$$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} e^{-x} dx \quad x = u^2.$$

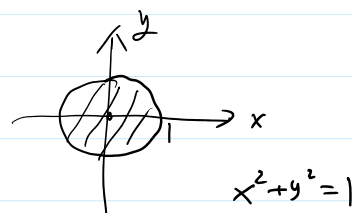
$$= 2 \int_0^{+\infty} u^{2\alpha-1} e^{-u^2} du \quad \alpha > 0.$$

$$\alpha = \frac{1}{2}.$$

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^{+\infty} e^{-u^2} du$$

= 重积分.

$$1. \text{ 几何: } f(x, y) \in C(D), \quad f(x, y) \geq 0.$$



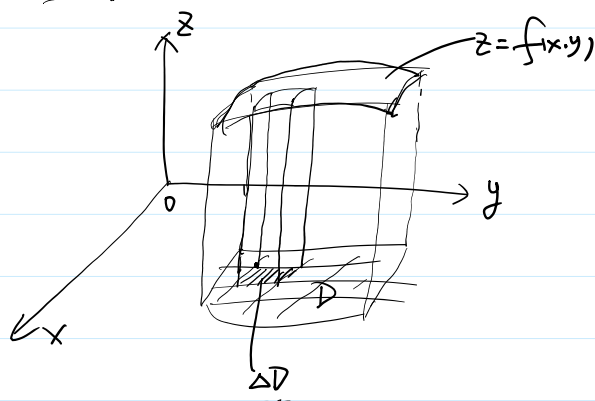
由区域 D, 以 T 为底

$\Sigma = \Gamma(x, y)$ 为 T 的

侧面积

$$x^2 + y^2 = 1$$

曲顶柱体: 以 D 为底. $z = f(x, y)$ 为顶. 母线平行于 z 轴.
作积.



$$\text{diam } \Delta D \rightarrow 0. \quad \forall (x, y) \in \Delta D.$$

ΔD 表示 ΔD 面积.

$$\sum_{i=1}^n V_i \approx \sum_{i=1}^n f(x_i, y_i) \cdot \Delta A_i$$

$$T = \{ \Delta D_1, \Delta D_2, \dots, \Delta D_n \}$$

$$(\Delta D_i)^\circ \cap (\Delta D_j)^\circ = \emptyset, \quad i \neq j.$$

$$D = \bigcup_{i=1}^n \Delta D_i$$

$$\|T\| = \{ \text{diam}(\Delta D_i) \mid i=1, 2, \dots, n \} \quad \text{—— 划分 } T \text{ 模.}$$

$$V = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta A_i$$

(2). 物理: 薄板上分布着某种物质, 分布不均匀. 求其质量.

$$\forall (x, y) \in D. \quad \rho(x, y) \text{ 面密度.}$$

$$T = \{ \Delta D_1, \dots, \Delta D_n \}.$$

$$\|T\| \rightarrow 0.$$

$$\Delta D_i \text{ 面积 } \Delta A_i.$$

$$\forall (x_i, y_i) \in \Delta D_i$$

$$M = \lim_{\|T\| \rightarrow 0} \sum_{i=1}^n \rho(x_i, y_i) \Delta A_i$$

定义 1. 设 $f(x, y)$ 定义在 $D \subset \mathbb{R}^2$ 上. D 有界闭区域.

$$\forall T = \{ \Delta D_1, \Delta D_2, \dots, \Delta D_n \} \quad \forall (x_i, y_i) \in \Delta D_i, \quad \Delta A_i \text{ 为 } \Delta D_i \text{ 面积.}$$

若 $\lim_{\|T\| \rightarrow 0} \sum_{i=1}^n f(x_i, y_i) \Delta A_i$ 存在. 且与划分 T 无关. 与点选取无关.

则 $f(x, y)$ 在 D 上可积. 且 $\iint_D f(x, y) da = \iint_{f(u, v)} f(u, v) da$ 简记 $\iint f da$

则 $f(x, y)$ 在 D 上可积. 且 $\iint_D f(x, y) da = \iint_D f(u, v) da$ 简记 $\iint_D f da$
 da — 面积微元.

①. $f(x, y), g(x, y) \in R(D)$.

$\forall \alpha, \beta \in \mathbb{R}. \quad \alpha f + \beta g \in R(D) \quad \text{且} \quad \iint_D (\alpha f + \beta g) da = \alpha \iint_D f da + \beta \iint_D g da$

②. $D = \bigcup_{i=1}^n D_i \quad D_i \cap D_j = \emptyset \quad (i \neq j).$

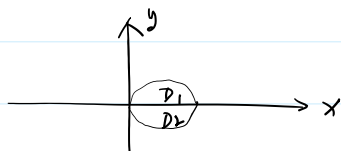
$f(x, y) \in R(D) \Leftrightarrow f(x, y) \in R(D_i). \quad \text{且} \quad \iint_D f da = \sum_{i=1}^n \iint_{D_i} f da$

③. 若 $f(x, y), g(x, y) \in R(D). \quad \text{且} \quad f(x, y) \leq g(x, y) \quad \forall (x, y) \in D.$

$$\iint_D f(x, y) da \leq \iint_D g(x, y) da$$

④ 若 $f(x, y) \in R(D). \quad \text{则} \quad \left| \iint_D f(x, y) da \right| \leq \iint_D |f(x, y)| da$

⑤. 若 $D = D_1 \cup D_2. \quad D_1, D_2$ 关于 x 轴对称. 且 $f(x, -y) = -f(x, y)$

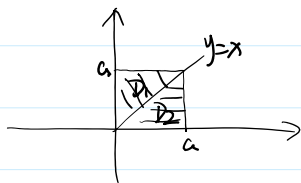


$$(x, y) \in D_1 \Leftrightarrow (x, -y) \in D_2$$

$$\begin{aligned} \iint_D f(x, y) da &= \iint_{D_1} f(x, y) da + \iint_{D_2} f(x, y) da \\ &= \iint_{D_2} f(x, -y) da + \iint_{D_2} f(x, y) da \\ &= 0. \end{aligned}$$

⑥. $f(x, -y) = f(x, y). \quad \text{则} \quad \iint_D f(x, y) da = 2 \iint_{D_1} f(x, y) da.$

⑥. (轮换对称性) 若 D 关于 $y=x$ 对称. 则 $\iint_D f(x, y) da = \iint_D f(y, x) da = \frac{1}{2} \iint_D (f(x, y) + f(y, x)) da$.



$$(x, y) \in D_1 \Leftrightarrow (y, x) \in D_2$$

$$\iint_{D_1} f(x, y) da + \iint_{D_2} f(x, y) da$$

$$\begin{aligned} &= \iint_{D_2} f(y, x) da + \iint_{D_1} f(y, x) da = \iint_D f(y, x) da \\ &= \iint_D f(x, y) da \end{aligned}$$